

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TRAINING TR-102 REPORT DAY 12

8 July 2025

Overview:

Today's session gave us a deep dive into how neural networks, the backbone of many AI systems, work. We learned about the structure of neural networks, how data flows through layers using forward propagation, and how the model learns using backward propagation (also called backpropagation). The session helped us understand how machines adjust their internal weights based on the errors they make — a key step in learning.

Learning Objectives:

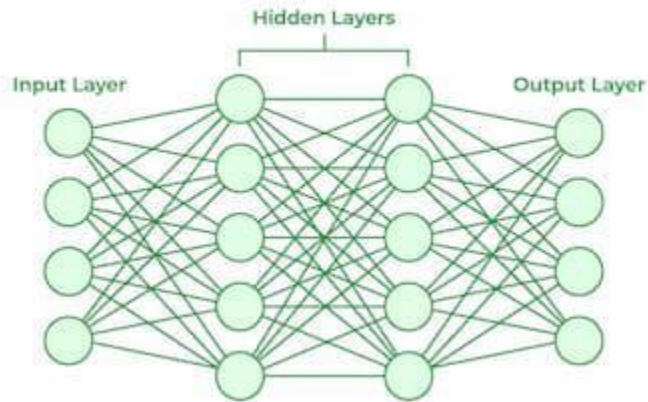
- Understand the basic structure and working of a neural network
- Learn how forward propagation processes input through layers
- Understand how backward propagation helps improve accuracy by adjusting weights
- See how loss functions and gradients play a key role in learning

Neural Networks:

A neural network is a collection of layers made of nodes that mimic the way a human brain works. These layers include:

- Input Layer: Where data enters
- Hidden Layers: Where processing happens
- Output Layer: Gives final result

Each connection has a weight, and each neuron uses an activation function to decide what to pass to the next layer.



Forward Propagation:

Forward propagation is the initial step in training a neural network, where the input data is fed through the network to generate a prediction.

Input Data: In the case of a feedforward neural network, the data flows in one direction — from input to output. The input layer consists of the data that you feed into the network. This could be any type of data suitable for your problem, like images, text, or numerical data. The data is often vectorized or turned into arrays that the network can process.

Weights and Bias: Weights and biases are the learnable parameters of a neural network. The weights determine the strength of the influence of a given input neuron on the output, while biases allow the output to be shifted by a constant value.

Weights: Each connection between neurons is assigned a weight, which is a measure of its relative importance. When an input is passed through a neuron, it gets multiplied by this weight. Weights can be both positive (excitatory) or negative (inhibitory), and they are initialized randomly at the start of training.

Bias: Bias is an extra input to neurons, and it is always 1, and has its own connection weight. This bias weight gives the ability to shift the activation function to the left or right, which may be critical for successful learning.

Neurons and Layers: The input data is fed into what we call neurons or nodes. These neurons are organized into multiple layers — an input layer, hidden layers, and an output layer.

Input Layer: This is where the network receives input from your dataset. It's often transformed into a suitable form for the network.

Hidden Layer(s): These are layers in between the input and output layers. They perform transformations on the input data with the goal of learning features that are useful for the task at hand. The term “deep” in deep learning refers to having multiple hidden layers in the network.

Output Layer: This is the final layer. It provides the predictions or classifications that the network has been trained to produce.

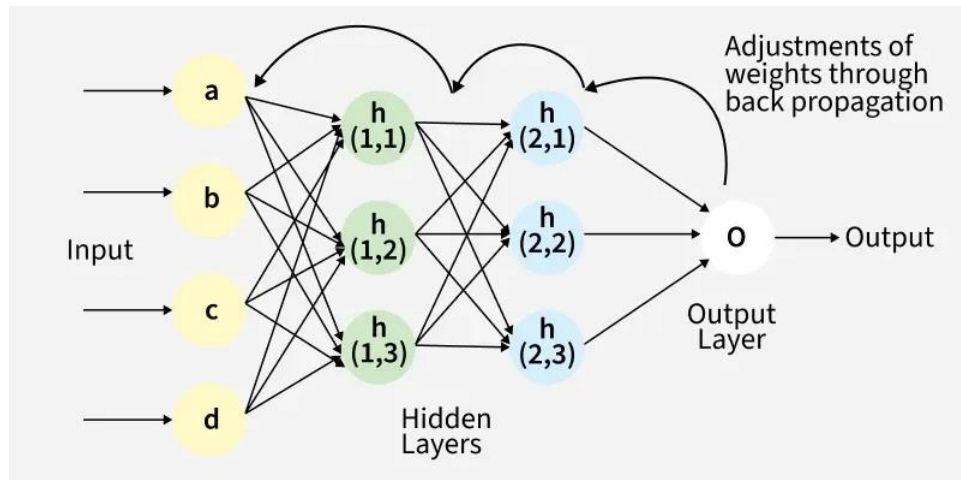
Linear Transformation: The first step in forward propagation involves calculating a weighted sum of the inputs and the associated weights and then adding the bias. This is also known as a linear transformation. The equation is usually as follows: $z = \text{weights} * \text{inputs} + \text{bias}$.

Backward Propagation:

Back Propagation is also known as "Backward Propagation of Errors" is a method used to train neural network. Its goal is to reduce the difference between the model's predicted output and the actual output by adjusting the weights and biases in the network.

Back Propagation plays a critical role in how neural networks improve over time.

- **Efficient Weight Update:** It computes the gradient of the loss function with respect to each weight using the chain rule making it possible to update weights efficiently.
- **Scalability:** The Back Propagation algorithm scales well to networks with multiple layers and complex architectures making deep learning feasible.
- **Automated Learning:** With Back Propagation the learning process becomes automated and the model can adjust itself to optimize its performance.



Conclusion:

Today's session helped us understand the core working of neural networks—how they predict and how they learn from mistakes. Forward propagation gave us insights into how data flows, while backward propagation taught us how the model improves through error correction. These steps are what make machine learning models smarter with time. It was a technical yet fascinating session that forms the base for deep learning concepts.